# LogWave: A Modern Ham Radio QSO Logging Solution for Amateur Radio Operators

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ABSTRACT
Amateur Radio operators require reliable, efficient, and accessible logging systems to maintain accurate records of radio contacts. While numerous commercial and open-source logging applications exist, many are either overly complex, resource-intensive, or focused on specialized operating environments. LogWave was developed to address these limitations by providing a lightweight, operator-centric logging solution that combines simplicity, usability, and modern data management practices. Written in Python and utilizing SQLite for database storage, LogWave provides contact logging, callsign lookup, search facilities, duplicate prevention, import/export capabilities, editable records, and visual operating aids. This paper describes the motivation, architecture, implementation, and operational capabilities of LogWave Version 1 and discusses its applicability to daily amateur radio operation.

**Keywords:** Amateur Radio, Ham Radio Logging, QSO Logger, Python, SQLite, Radio Communication, Log Management, Software Development.




**Figure 1:** Main operating screen of LogWave .

## 1.INTRODUCTION

Logging has been an integral part of amateur radio operation since the earliest days of wireless communication. Maintaining an accurate logbook serves multiple purposes, including operating records, award verification, contest participation, QSL management, and station analysis.

Historically, radio amateurs maintained handwritten logs. With the advent of personal computers, electronic logging systems became commonplace. While modern logging software offers powerful capabilities, many applications require extensive configuration and training before they can be used effectively.

The objective behind the development of LogWave was to create a practical logging application that emphasizes speed, simplicity, readability, and operator workflow. Rather than existing logging programs often provide extensive features but may require significant configuration and internet connectivity before daily operation. overwhelming operators with complex configuration menus and specialized functions, As an active amateur radio operator, I wanted a logging application that was simple, fast, readable, and optimized for routine station operation. LogWave was developed to address these requirements and focuses on features that amateurs regularly use during normal station operation.

**2. DESIGN OBJECTIVES**

The development of LogWave was guided by five primary design objectives:

**A. Simplicity**

The software should allow contact information to be entered quickly with minimal operator interaction.

**B. Readability**

Operator fatigue can become a significant issue during long operating sessions. Therefore, special attention was given to font sizes, color contrast, and control placement.

**C. Reliability**

The software must operate reliably without requiring internet connectivity, cloud services, or external database systems.

**D. Portability**

The logging database should be easily transferred between computers without specialized installation procedures.

**E. Expandability**

Future capability enhancements should be possible without major architectural redesign.

**3. SYSTEM ARCHITECTURE**

LogWave is a windows based application implemented using Python with a layered software architecture.

The application consists of four major components:

**A. User Interface Layer**

The graphical user interface was implemented using Tkinter and ttk widgets.

Responsibilities include:

- Callsign entry
- QSO management
- Search operations
- Record editing
- Status monitoring

**B. Database Layer**

SQLite was selected as the database engine because of its:

- Lightweight design
- Reliability
- Portability
- Low maintenance requirements

All contact records are stored in a single local database file.

**C. Data Management Layer**

This layer handles:

- Contact storage
- Duplicate detection
- Search operations
- Sorting functions
- Statistics generation

**D. Import/Export Layer**

Data exchange is supported using:

- Excel (.xlsx)
- Legacy Excel (.xls)
- CSV (.csv)

This ensures compatibility with historical logging data and external applications.

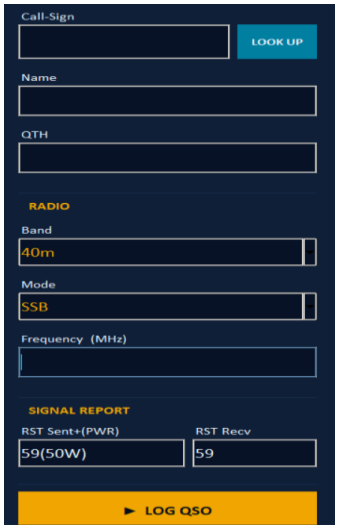



**Figure 2** Contact entry panel illustrating operator-centric workflow.

**4. USER INTERFACE DESIGN**

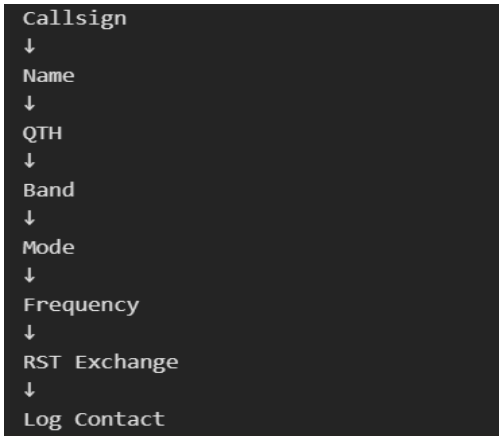
A major focus of LogWave is operator-centered interface design.

The application window consists of two primary sections:

1. Contact Entry Panel
2. QSO Display Panel

**Contact Entry Workflow**

The input sequence follows the natural order used during radio contacts:




This arrangement reduces operator workload and minimizes data entry errors.

**5. CALLSIGN LOOKUP SYSTEM**

The Callsign Lookup function was developed to reduce repetitive data entry.

When the operator enters a callsign and activates the lookup function, LogWave executes the following sequence:

1. Search the local database.
2. Retrieve previously stored Name and QTH information.
3. Query online lookup services if local records are unavailable.

This process significantly improves logging speed while maintaining consistency.




**Figure 3:** Automatic population of Name and QTH fields after execution of a callsign lookup operation

**6. QSO DATABASE MANAGEMENT**

All contacts are displayed in a centralized database view.

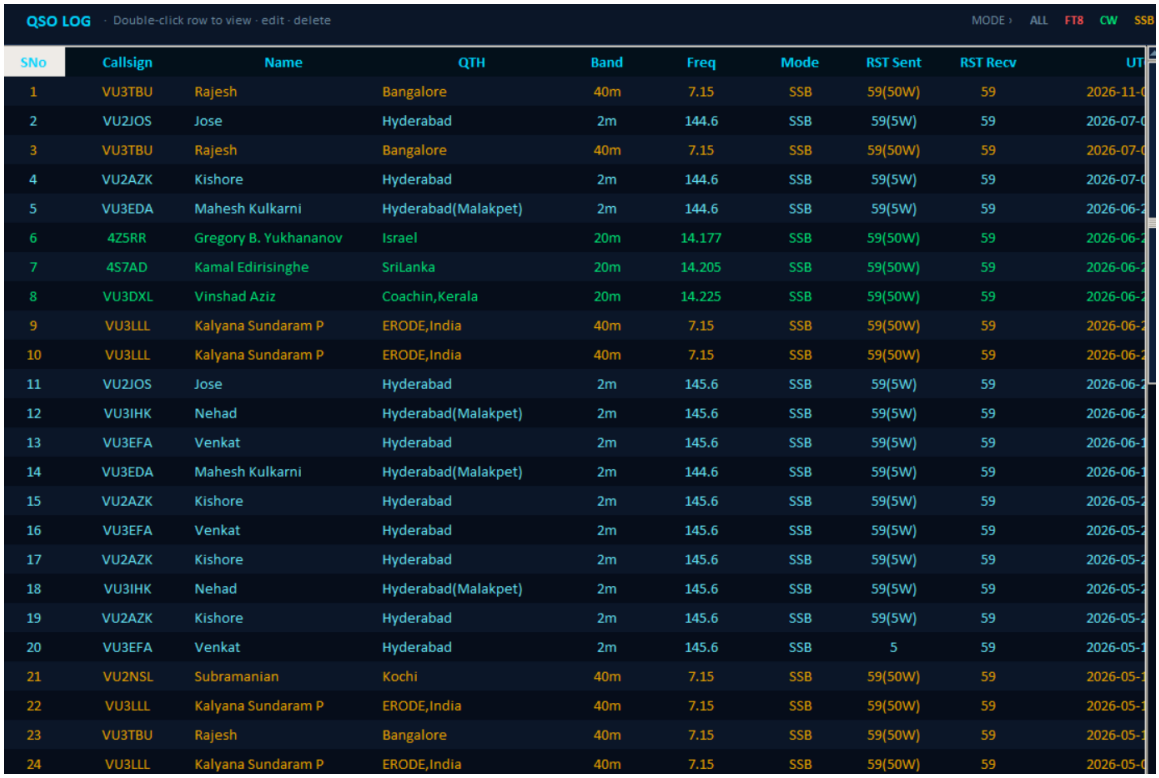
The displayed fields include:

| Field | Description |
|---|---|
| SNo | Serial Number |
| Callsign | Station Worked |
| Name | Operator Name |

| | |
|---|---|
| QTH | Station Location |
| Band | Operating Band |
| Frequency | Operating Frequency |
| Mode | Communication Mode |
| RST Sent | Signal Report Sent |
| RST Received | Signal Report Received |
| UTC Time | Universal Coordinated Time |

The display supports both vertical and horizontal scrolling, enabling efficient navigation of large logging databases.

The database view also supports column-based sorting.




| SNo | Callsign | Name | QTH | Band | Freq | Mode | RST Sent | RST Recv | UT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | VU3TBU | Rajesh | Bangalore | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-11-0 |
| 2 | VU2JOS | Jose | Hyderabad | 2m | 144.6 | SSB | 59(5W) | 59 | 2026-07-0 |
| 3 | VU3TBU | Rajesh | Bangalore | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-07-0 |
| 4 | VU2AZK | Kishore | Hyderabad | 2m | 144.6 | SSB | 59(5W) | 59 | 2026-07-0 |
| 5 | VU3EDA | Mahesh Kulkarni | Hyderabad(Malakpet) | 2m | 144.6 | SSB | 59(5W) | 59 | 2026-06-2 |
| 6 | 4Z5RR | Gregory B. Yukhananov | Israel | 20m | 14.177 | SSB | 59(50W) | 59 | 2026-06-2 |
| 7 | 4S7AD | Kamal Edirisinghe | SriLanka | 20m | 14.205 | SSB | 59(50W) | 59 | 2026-06-2 |
| 8 | VU3DXL | Vinshad Aziz | Coachin,Kerala | 20m | 14.225 | SSB | 59(50W) | 59 | 2026-06-2 |
| 9 | VU3LLL | Kalyana Sundaram P | ERODE,India | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-06-2 |
| 10 | VU3LLL | Kalyana Sundaram P | ERODE,India | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-06-2 |
| 11 | VU2JOS | Jose | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-06-2 |
| 12 | VU3IHK | Nehad | Hyderabad(Malakpet) | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-06-2 |
| 13 | VU3EFA | Venkat | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-06-1 |
| 14 | VU3EDA | Mahesh Kulkarni | Hyderabad(Malakpet) | 2m | 144.6 | SSB | 59(5W) | 59 | 2026-06-1 |
| 15 | VU2AZK | Kishore | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-05-2 |
| 16 | VU3EFA | Venkat | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-05-2 |
| 17 | VU2AZK | Kishore | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-05-2 |
| 18 | VU3IHK | Nehad | Hyderabad(Malakpet) | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-05-2 |
| 19 | VU2AZK | Kishore | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-05-2 |
| 20 | VU3EFA | Venkat | Hyderabad | 2m | 145.6 | SSB | 5 | 59 | 2026-05-1 |
| 21 | VU2NSL | Subramanian | Kochi | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-05-1 |
| 22 | VU3LLL | Kalyana Sundaram P | ERODE,India | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-05-1 |
| 23 | VU3TBU | Rajesh | Bangalore | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-05-1 |
| 24 | VU3LLL | Kalyana Sundaram P | ERODE,India | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-05-0 |

Sortable QSO database display.

**7. VISUAL IDENTIFICATION FEATURES**

LogWave employs visual enhancements designed to improve readability during active operation.

**A. Colour-Coded Bands**

Bands are assigned unique colours, enabling rapid visual recognition of operating frequency ranges.

**B. Colour-Coded Modes**

Operating modes such as SSB, CW, FT8, and FM are assigned distinctive colours.

**C. Dark Theme Interface**

The software employs a dark-color operating environment to reduce eye strain during extended operation.

**D. Large Fonts and Controls**

High-visibility controls improve readability on modern high-resolution displays.

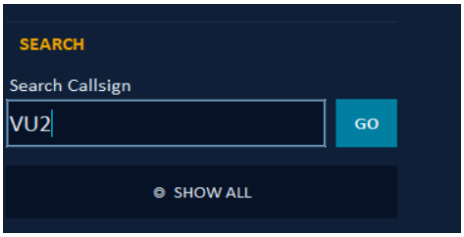
## 8. SEARCH AND DATA RETRIEVAL

As log databases grow, rapid retrieval of historical records becomes increasingly important.

The integrated search system allows operators to filter contacts based on partial or complete callsigns.

For example:

VU2




returns all stations matching the entered pattern.

QSO LOG · Double-click row to view · edit · delete

MODE › ALL FT8 CW

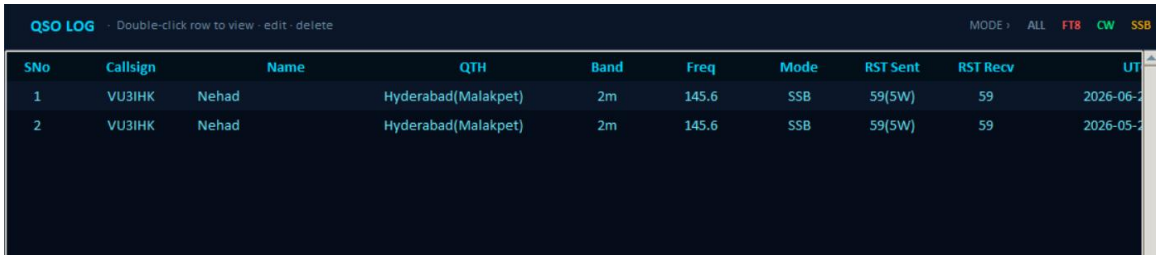
| SNo | Callsign | Name | QTH | Band | Freq | Mode | RST Sent | RST Recv | UT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | VU2JOS | Jose | Hyderabad | 2m | 144.6 | SSB | 59(5W) | 59 | 2026-07- |
| 2 | VU2AZK | Kishore | Hyderabad | 2m | 144.6 | SSB | 59(5W) | 59 | 2026-07- |
| 3 | VU2JOS | Jose | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-06- |
| 4 | VU2AZK | Kishore | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-05- |
| 5 | VU2AZK | Kishore | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-05- |
| 6 | VU2AZK | Kishore | Hyderabad | 2m | 145.6 | SSB | 59(5W) | 59 | 2026-05- |
| 7 | VU2NSL | Subramanian | Kochi | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-05- |
| 8 | VU2UCR | Hari | Bengalore,KA | 20m | 14.175 | SSB | 59(50W) | 59 | 2026-05- |
| 9 | VU2KOC | OM Prakash | Belgum,karnatak,Ind | 40m | 7.15 | SSB | 59(50W) | 59 | 2026-05- |
| 10 | VU2DSI | Datta | Ahmednagar | 20m | 14.215 | SSB | 59 | 59 | 2026-05- |
| 11 | VU2SRB | Sudheer | Nagpur | 40m | 14.235 | SSB | 59 | 59 | 2026-05- |
| 12 | VU2SNK | Satish | Nasik | 20m | 14.235 | SSB | 59 | 59 | 2026-05- |
| 13 | VU2KOC | OM Prakash | Belgum,karnatak,Ind | 40m | 7.15 | SSB | 59 | 59 | 2026-05- |
| 14 | VU2RCT | Chandra | Kavoor, MANGALORE | 20m | 14.218 | SSB | 59 | 59 | 2026-04- |
| 15 | VU2FRD | RAJESH KUMAR | BHUBANESWAR | 20m | 14.245 | SSB | 59 | 59 | 2026-04- |
| 16 | VU2RBI | D.Bharathi | Saidabad,Hyd | 20m | 14.2 | SSB | 59 | 59 | 2026-04- |
| 17 | VU2EVQ | ANEES P.I | MALAPPURAM,Kerala | 40m | 7.105 | SSB | 59 | 59 | 2026-04- |
| 18 | VU2YK | RAHUL KAPOOR | NEW DELHI | 20m | 14.215 | SSB | 59 | 59 | 2026-04- |
| 19 | VU2VID | VINOD . K | AYOOR, KERALA, | 20m | 14.195 | SSB | 59 | 59 | 2026-04- |
| 20 | VU2DSI | Dattatry Deogaonkar | Ahmednagar | 20m | 14.215 | SSB | 59 | 59 | 2026-04- |

Similarly:

Call sign search




returns all matching stations beginning with the specified prefix.




This functionality is especially valuable during:

- Award verification
- Contest operation
- Repeat contacts
- QSL confirmation

**9. DATA INTEGRITY AND DUPLICATE PREVENTION**

Database integrity was a key design consideration.

LogWave prevents accidental duplication using a unique combination of: Callsign + UTC Timestamp

During software startup, duplicate verification is executed automatically and duplicate entries are removed while preserving legitimate records.

This mechanism ensures:

- Consistent award tracking
- Accurate station records
- Reliable export data

**10. IMPORT AND EXPORT CAPABILITIES**

The system provides bidirectional data exchange functionality.

**Import Formats**

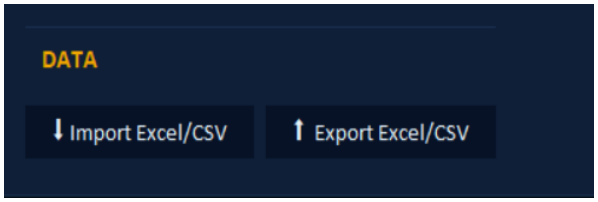
- XLSX
- XLS
- CSV

**Export Formats**

- XLSX
- CSV

These capabilities simplify:

- Data migration
- Backup creation
- Award processing
- Third-party analysis




| A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|
| callsign | band | mode | frequency | rst_sent | rst_recv | qth | name | utc_datetime |
| S79VU | 20m | SSB | 14.2 | 59 | 59 | Seychelles | Ravi Shanker Pandit | 29-04-2026 14:12 |
| VU2RCT | 20m | SSB | 14.218 | 59 | 59 | Kavoor, MANGALORE | Chandra | 29-04-2026 13:35 |
| VU3FWG | 20m | SSB | 14.2 | 59 | 59 | Kundali,Odissa | Debasis Mallik | 26-04-2026 15:46 |
| VU2FRD | 20m | SSB | 14.245 | 59 | 59 | BHUBANESWAR | RAJESH KUMAR | 26-04-2026 12:27 |
| VU2RBI | 20m | SSB | 14.2 | 59 | 59 | Saidabad,Hyd | D.Bharathi | 26-04-2026 11:49 |
| YB5DDE | 15m | SSB | 21.283 | 59 | 59 | Indonesia | SUHARDEDI ( DEDI ) | 25-04-2026 14:50 |
| VU2EVQ | 40m | SSB | 7.105 | 59 | 59 | MALAPPURAM,Kerala | ANEES P.I | 25-04-2026 14:43 |
| VU3LLL | 40m | SSB | 7.15 | 59 | 59 | ERODE,India | Kalyana Sundaram P | 25-04-2026 14:27 |
| YB1AAE | 15m | SSB | 21.251 | 55 | 52 | Jakarta, Indonesia | Muhammad Mansur 'INDAR' | 25-04-2026 13:23 |
| S79VU | 20m | SSB | 14.2 | 59 | 59 | Seychelles | Ravi Shanker Pandit | 23-04-2026 15:11 |
| S21PL | 20m | SSB | 14.215 | 59 | 59 | Bangladesh | Md. Pappu Mia | 23-04-2026 15:03 |
| 4S7AB | 20m | SSB | 14.19 | 59 | 59 | Sri Lanka | KAMAL EDIRISINGHE | 11-04-2026 13:55 |
| VU3BXI | 20m | SSB | 14.215 | 59 | 59 | Dwarka, New Delhi | Kaizer Ahamed | 04-04-2026 13:22 |
| VU2YK | 20m | SSB | 14.215 | 59 | 59 | NEW DELHI | RAHUL KAPOOR | 04-04-2026 13:18 |
| VU3DOD | 20m | SSB | 14.235 | 59 | 59 | Mitraon, Delhi | Devendra Kalonia | 04-04-2026 13:14 |
| VU32GO | 20m | SSB | 14.255 | 59 | 59 | Pune | Indrajit Bhat | 04-04-2026 13:04 |
| VU2VID | 20m | SSB | 14.195 | 59 | 59 | AYOOR, KERALA, | VINOD . K | 04-04-2026 12:55 |
| VU2DSI | 20m | SSB | 14.215 | 59 | 59 | Ahmednagar | Dattatry Deogaonkar | 04-04-2026 12:44 |
| VU3OII | 20m | SSB | 14.2 | 59 | 59 | PARAGANAS,IND | Sahir Alam Mondal | 04-04-2026 12:29 |
| BY7WZ | 20m | SSB | 14.23 | 59 | 59 | China | Wuzhou Amateur Radio | 29-03-2026 15:33 |

**Figure 7:** Excel and CSV import/export facilities used for data migration and station backup.

## 11. RECORD EDITING AND MAINTENANCE

Double-clicking a contact record opens the editing interface.

The operator may modify:

- Callsign
- Name
- QTH
- Band
- Mode
- Frequency
- RST Reports
- UTC Timestamp

The system also supports deletion of unwanted records.

This functionality ensures that errors can be corrected without requiring direct database access.

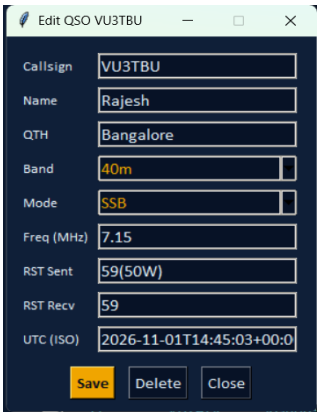



**Figure 8:** Contact editing interface allowing modification of stored QSO information.

## 12. PERFORMANCE EVALUATION

LogWave was evaluated under normal operating conditions using local databases containing several records.

Performance observations include:

- Fast application startup
- Minimal memory consumption
- Responsive search performance
- Reliable database operation
- Stable long-duration execution

Because SQLite is embedded within the application, no external database infrastructure is required.

The application remains fully functional without internet connectivity.




**Figure 9:** Real-time station statistics displaying total QSOs, unique callsigns worked, and operating band activity.

## 13. FUTURE DEVELOPMENT

Several enhancements are being considered for future versions of LogWave:

**A. ADIF Support**
Support for industry-standard Amateur Data Interchange Format files.
**B. QRZ Integration**
Automated retrieval of callsign information.
**C. LoTW Synchronization**
Direct integration with ARRL Logbook of The World.
**D. eQSL Integration**

Electronic QSL exchange and confirmation.

**E. Contest Mode**

Dedicated high-speed logging mode optimized for amateur radio contests.

**F. Award Tracking**

Automated monitoring of:

- DXCC
- WAS
- WAZ
- VU Awards

## 14. CONCLUSION

LogWave demonstrates that effective amateur radio logging software can be both powerful and simple. By emphasizing operator workflow, readability, data integrity, and ease of use, the application provides a practical solution for daily amateur radio operations.

The software integrates contact logging, callsign lookup, duplicate prevention, editable records, import/export capabilities, and visual operating aids within a lightweight and portable platform.

As the software continues to evolve, LogWave is expected to support a broader range of operating activities while maintaining its core philosophy of simplicity and usability.

LogWave Version 1.0 has been developed as a personal project for the Amateur Radio community. The source code is available free of charge for educational and amateur radio use. I also I welcome feedback and suggestions from the Amateur Radio community. Operators interested in evaluating, modifying, or contributing to LogWave and Interested operators may request a copy by contacting the author via email at **vu3bpq@gmail.com** or The source code is available on GitHub at free of charge to interested amateur radio operators https://github.com/vu3bpq-ArunChutke/LogWave.git.

**73 de VU3BPQ**
**LogWave Version 1.0**
**Radio Logging Made Easy**